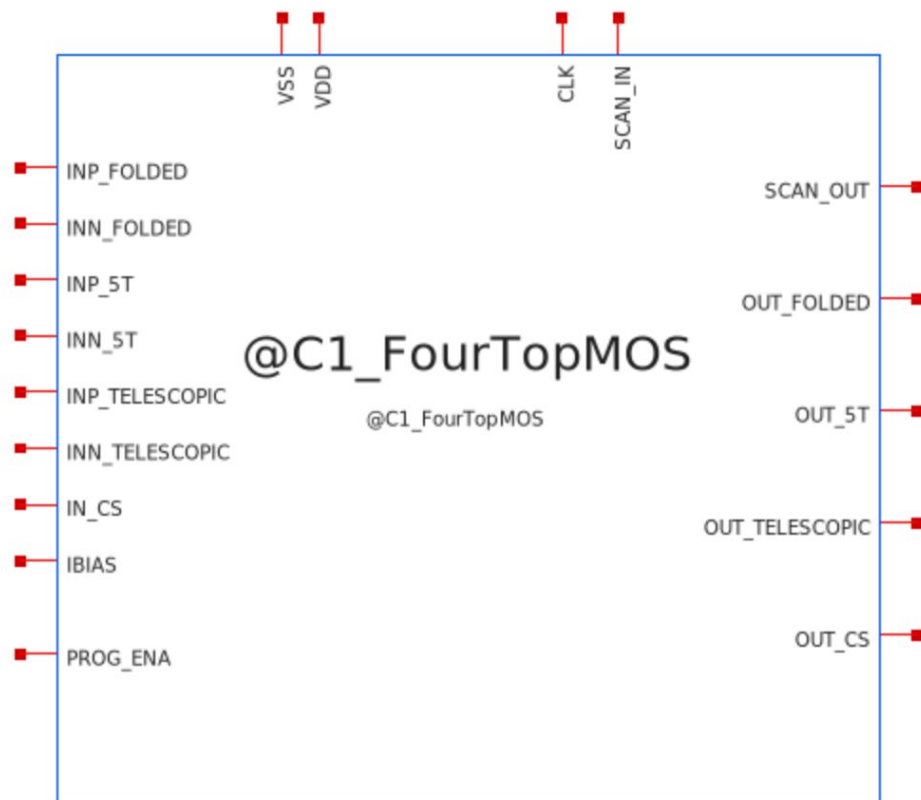


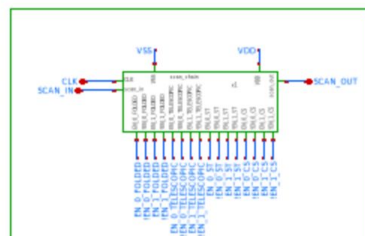
# C1 Four Topology MOSbius

## Four Topology MOSbius

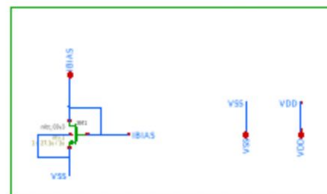
| Discord name     | Affiliation         | Experience    | Role        |
|------------------|---------------------|---------------|-------------|
| elij0100         | Columbia University | Undergraduate | Team Lead   |
| boibun.          | Columbia University | Undergraduate | Team Member |
| maxdrucker_83950 | Columbia University | Undergraduate | Team Member |
| Manwell          | Columbia University | Undergraduate | Team Member |



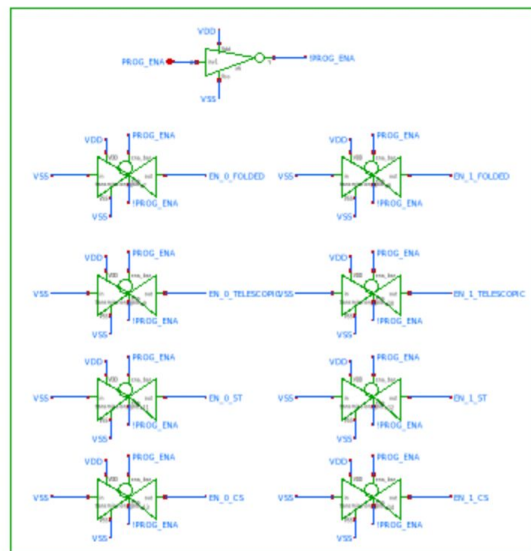
## SCAN CHAIN



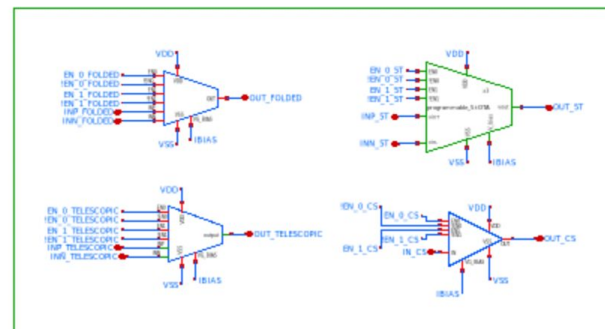
## BIAS AND POWER PINS



## PROGRAMMING ENABLE



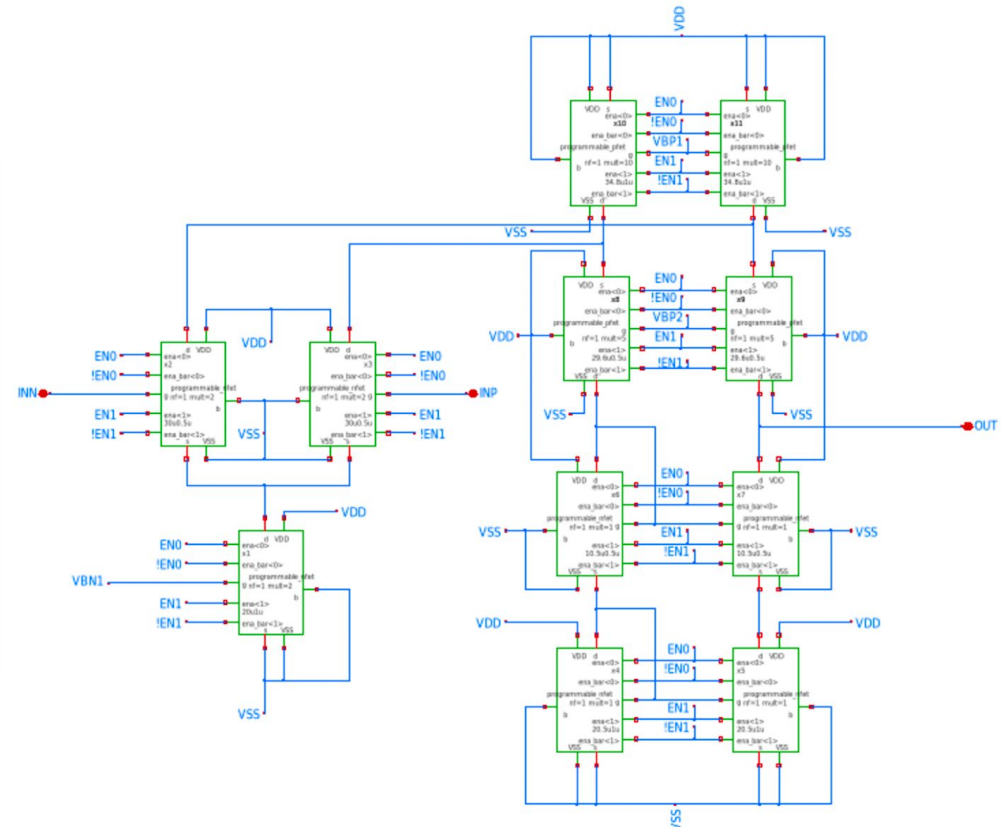
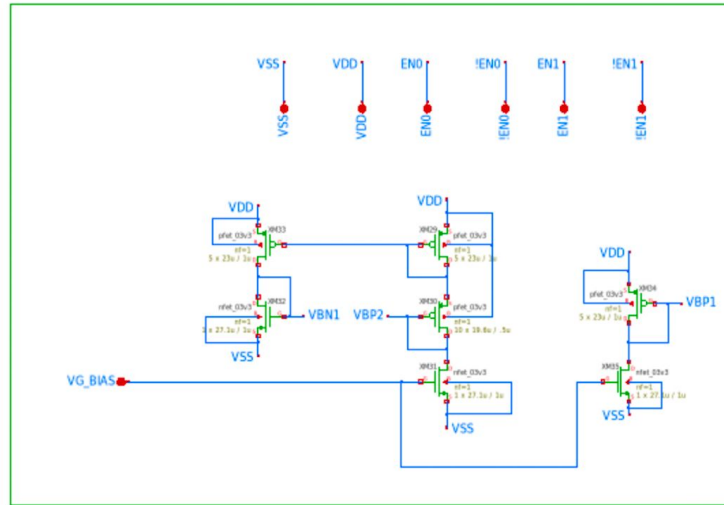
## CORE AMPLIFIERS



Pull Downs on  
Enable Pins



## Bias Voltage Generation and Pins



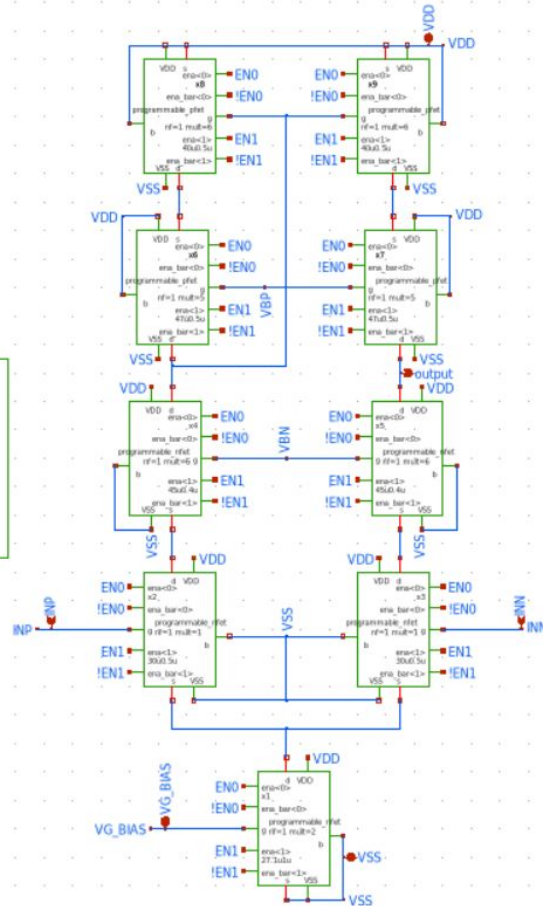
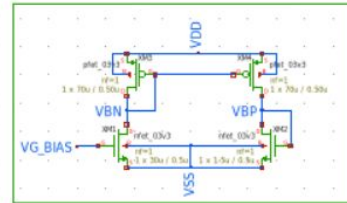
# Folded Cascode Sizing

| Device    | Role                 | W ( $\mu\text{m}$ ) | L ( $\mu\text{m}$ ) | ID ( $\mu\text{A}$ ) |
|-----------|----------------------|---------------------|---------------------|----------------------|
| XM10/XM11 | NMOS input pair      | 60                  | 0.5                 | 75                   |
| XM12      | NMOS tail            | 40                  | 1                   | 150                  |
| XM2/XM5   | PMOS current sources | 174                 | 1                   | 150                  |
| XM3/XM4   | PMOS cascodes        | 148                 | 0.5                 | 75                   |
| XM6/XM7   | NMOS cascodes        | 10.5                | 0.5                 | 75                   |
| XM8/XM9   | NMOS mirrors         | 20.5                | 1                   | 75                   |

- (Default sizing shown, for 2x, 3x, and 4x sizing, the width and bias current would scale respectively for all amplifiers)

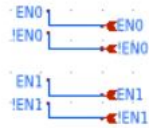
EN0  
!EN0  
EN1  
!EN1

### Bias Voltage Circuit



# Telescopic Cascode Sizing

| Device  | Role                    | W ( $\mu\text{m}$ ) | L ( $\mu\text{m}$ ) | ID ( $\mu\text{A}$ ) |
|---------|-------------------------|---------------------|---------------------|----------------------|
| XM1     | NMOS Tail               | 60                  | 0.5                 | 200                  |
| XM2/XM3 | Input Pair              | 30                  | 0.5                 | 100                  |
| XM4/XM5 | NMOS<br>Cascode         | 180                 | 0.5                 | 100                  |
| XM6/XM7 | PMOS<br>Sascodes        | 235                 | 0.4                 | 100                  |
| XM8/XM9 | PMOS Current<br>Sources | 240                 | 0.5                 | 100                  |

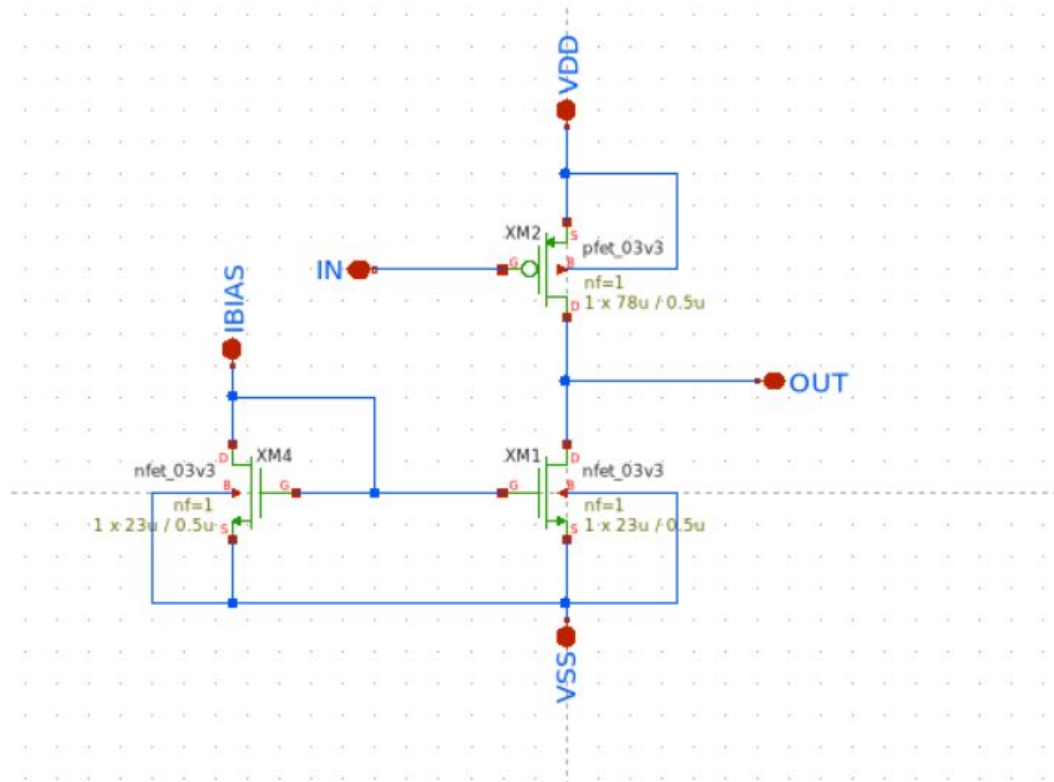




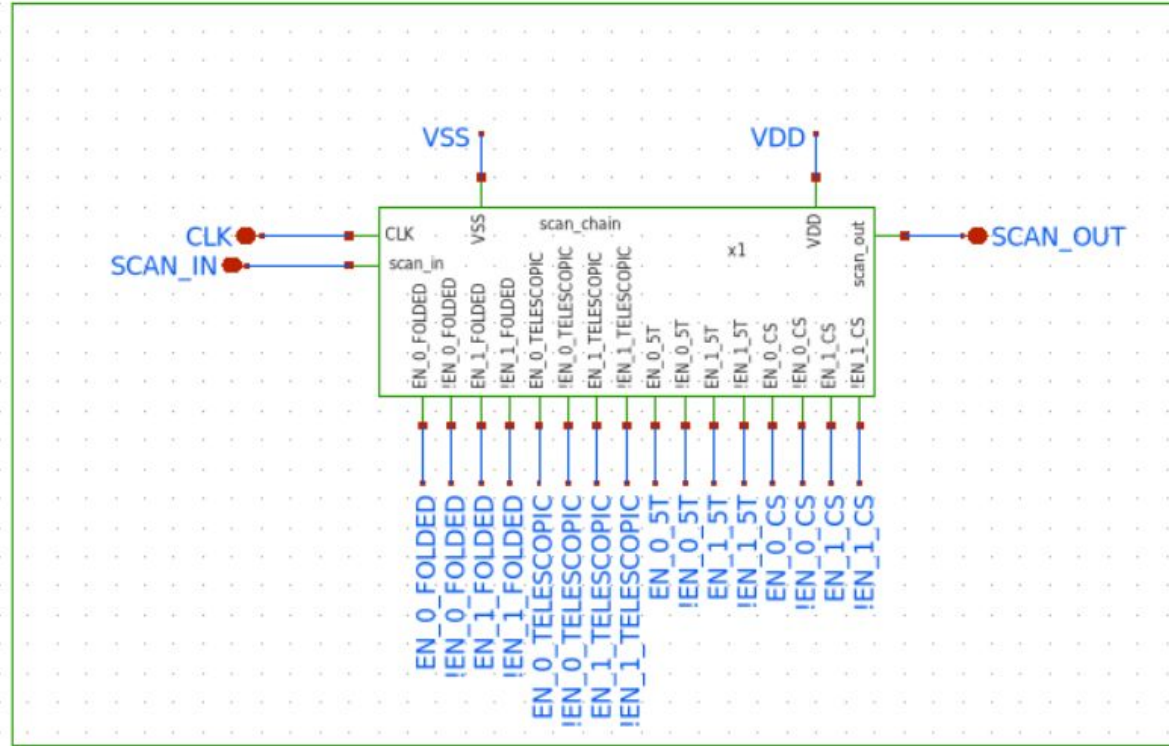
## 5-Transistor Sizing

| Device  | Role             | W ( $\mu\text{m}$ ) | L ( $\mu\text{m}$ ) | ID ( $\mu\text{A}$ ) |
|---------|------------------|---------------------|---------------------|----------------------|
| XM1/XM2 | Input Pair       | 34.2                | 0.5                 | 150                  |
| XM3/XM4 | PMOS Active Load | 100                 | 0.5                 | 150                  |
| XM5     | NMOS Tail        | 68.4                | 0.5                 | 300                  |

# Common-Source Amplifier



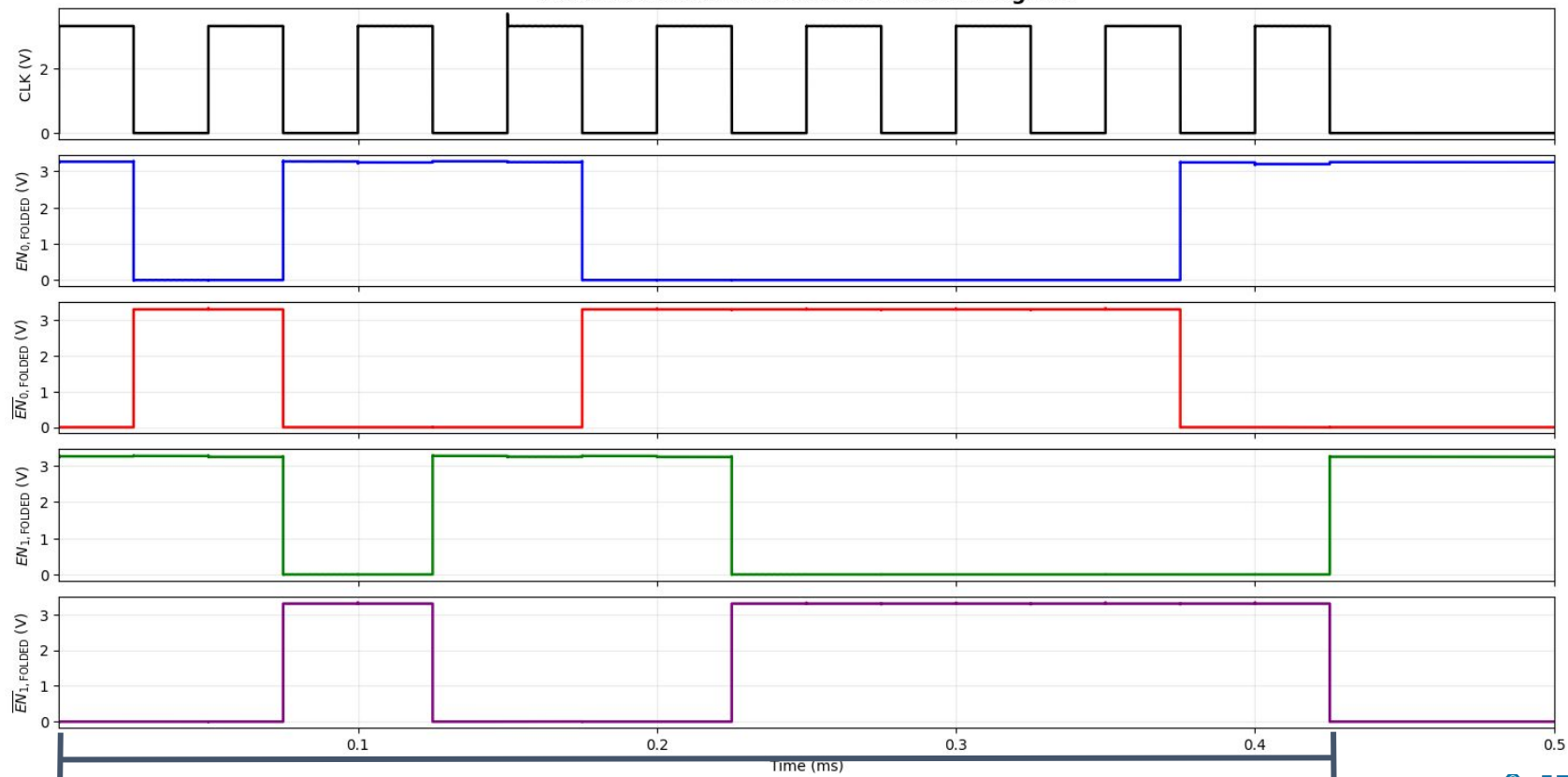
# SCAN CHAIN



Clock and Scan-In Signal are to be generated by a Raspberry Pi running micropython (can also be replaced by any other microcontroller). Data for any possible configuration can be propagated in .425 mS, and verified via the Scan-Out Pin. In the next slide, the D Flip-Flop outputs of EN0 and EN1 (and compliments) are displayed, propagated in .425 mS. At .425 mS the clock turns off, and with the enable bits set in place until the user chooses to reconfigure the Four Topology MOSbius.

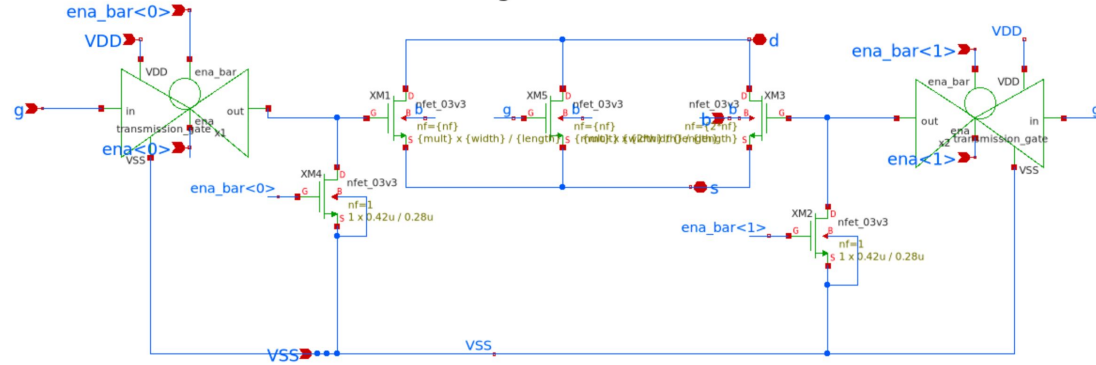
\*During this data propagation, in transient simulations, you will notice voltage spikes do to digital switching, which disappear after the clock is turned off.

Folded Cascode OTA Transient Control Signals

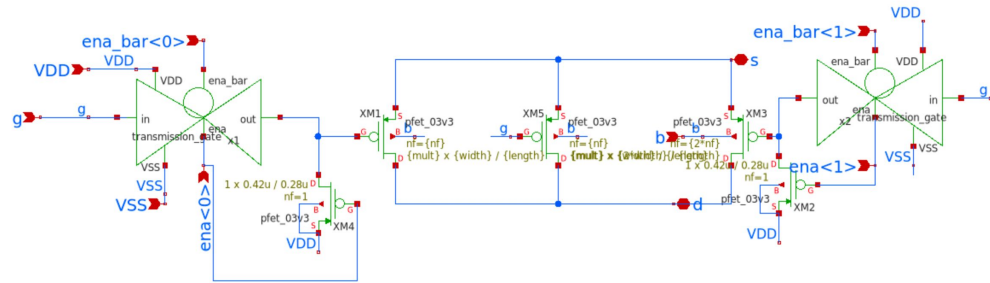


.425 mS Data Propagation

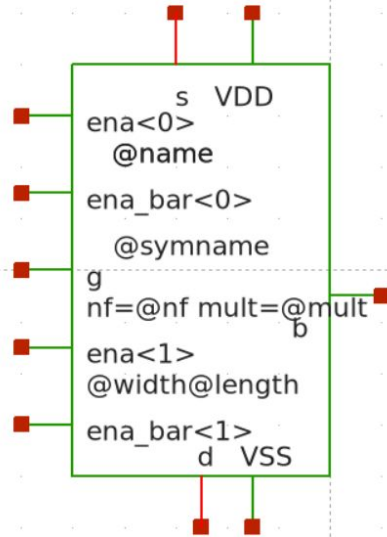
## Programmable NFET



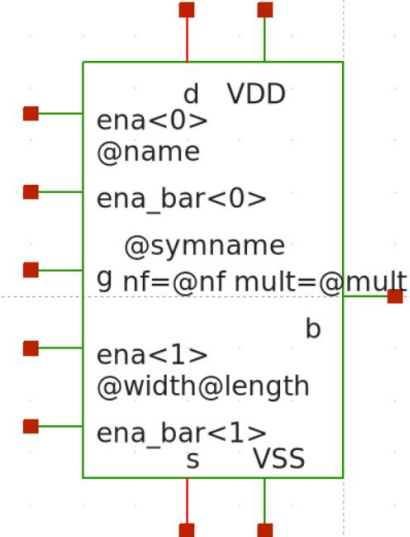
## Programmable PFET



# Programmable PFET



# Programmable NFET

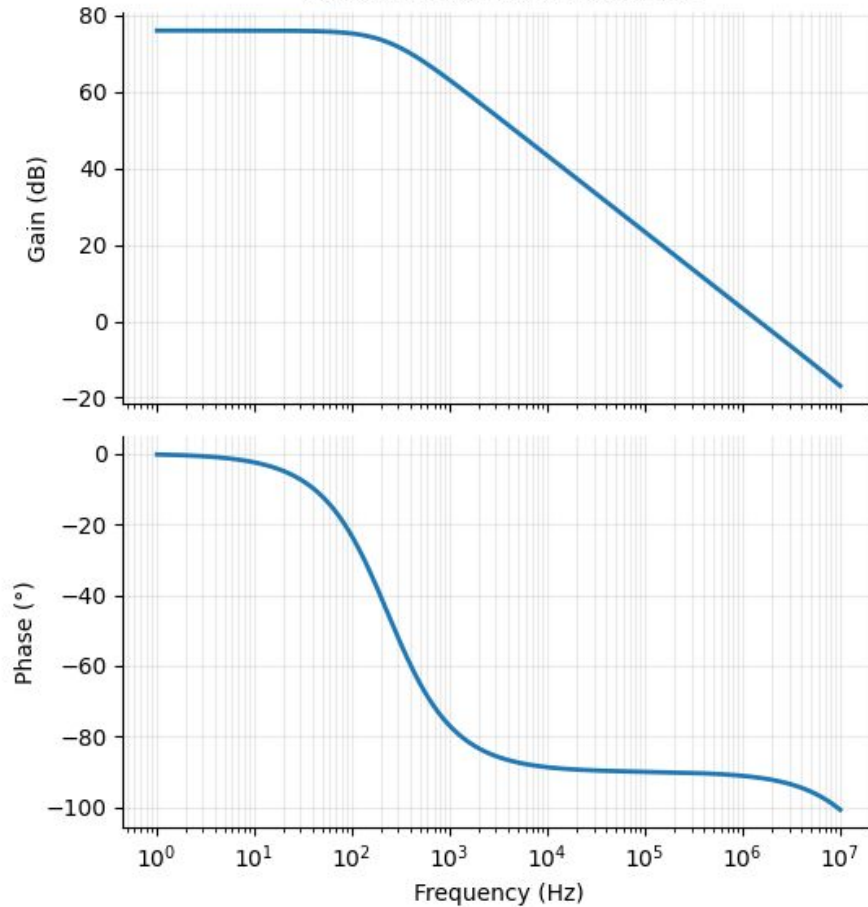


## AC Simulations

The following AC Sims were extracted by hard coding the Programming Transistors (Setting the Enable Pins high or low without the need of the Scan Chain). AC sims were unable to be done directly via the Top-Level as SPICE was unable to find an operation point due to the presence of the digital platform. Instead, GBW was extracted via the GBW product. There is an error margin as result were limited by 1uS resolution simulations. These results can be found in the last slide.

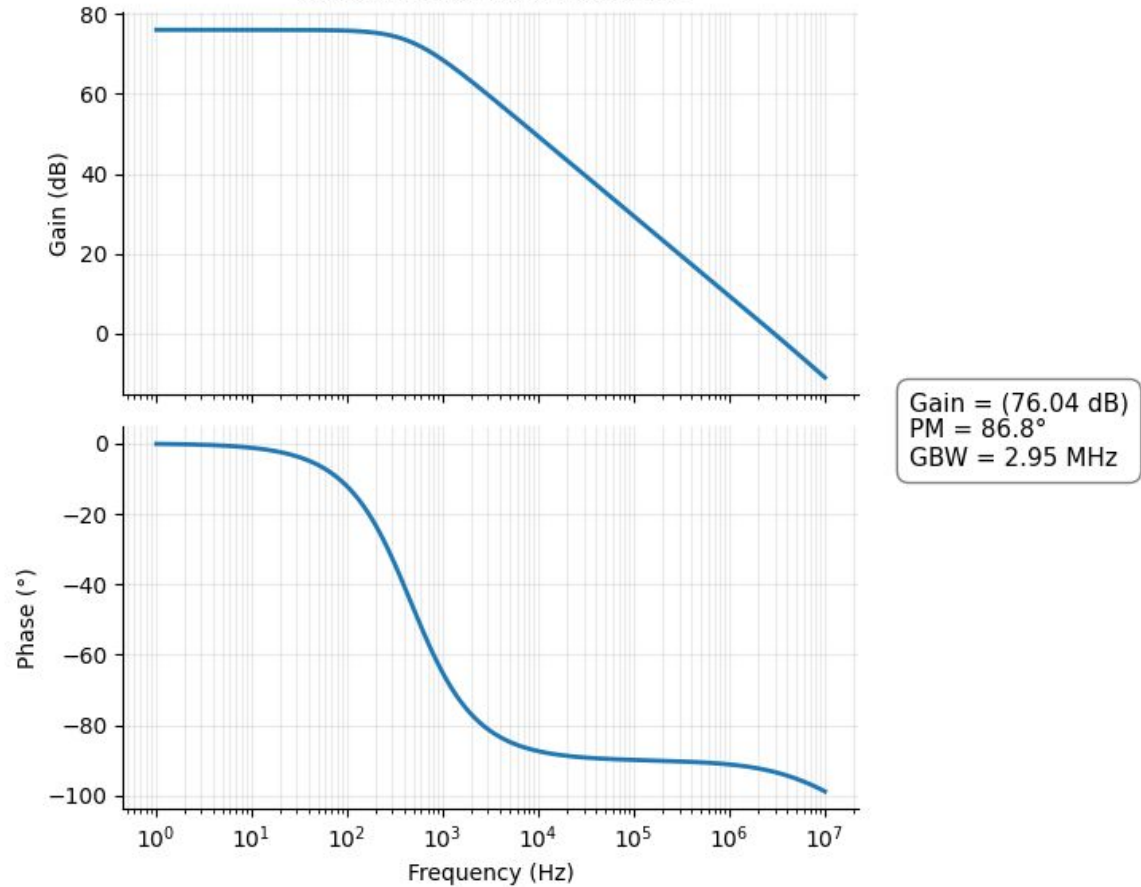


1X Folded Cascode Bode Plot

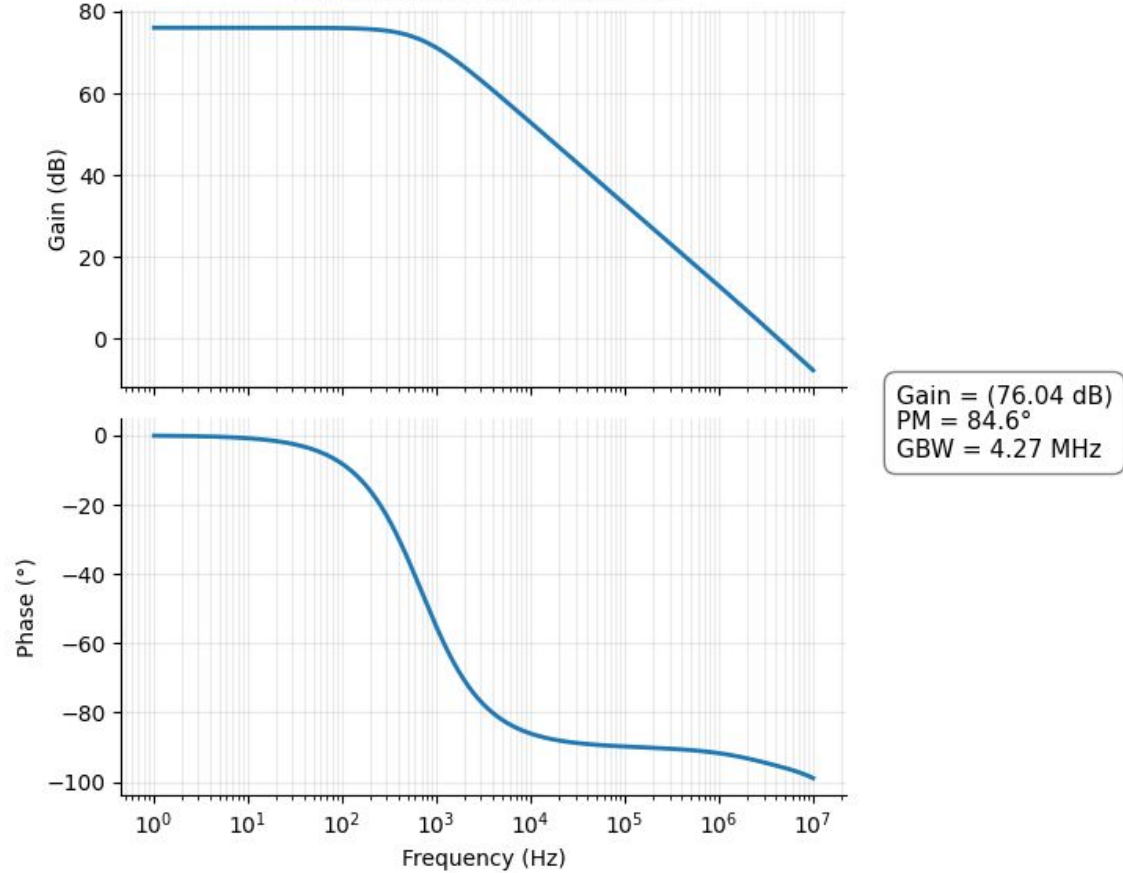


Gain = (76.04 dB)  
PM = 88.3°  
GBW = 1.51 MHz

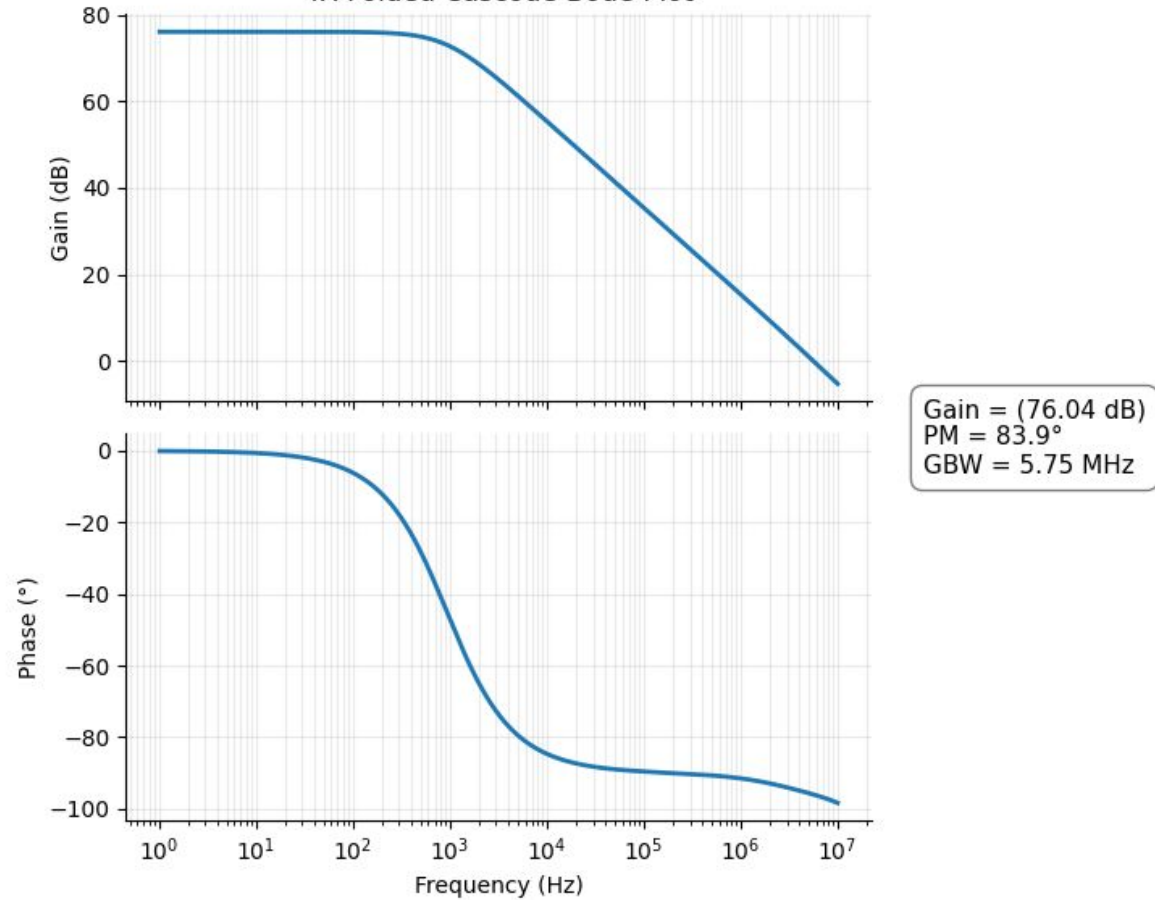
2X Folded Cascode Bode Plot



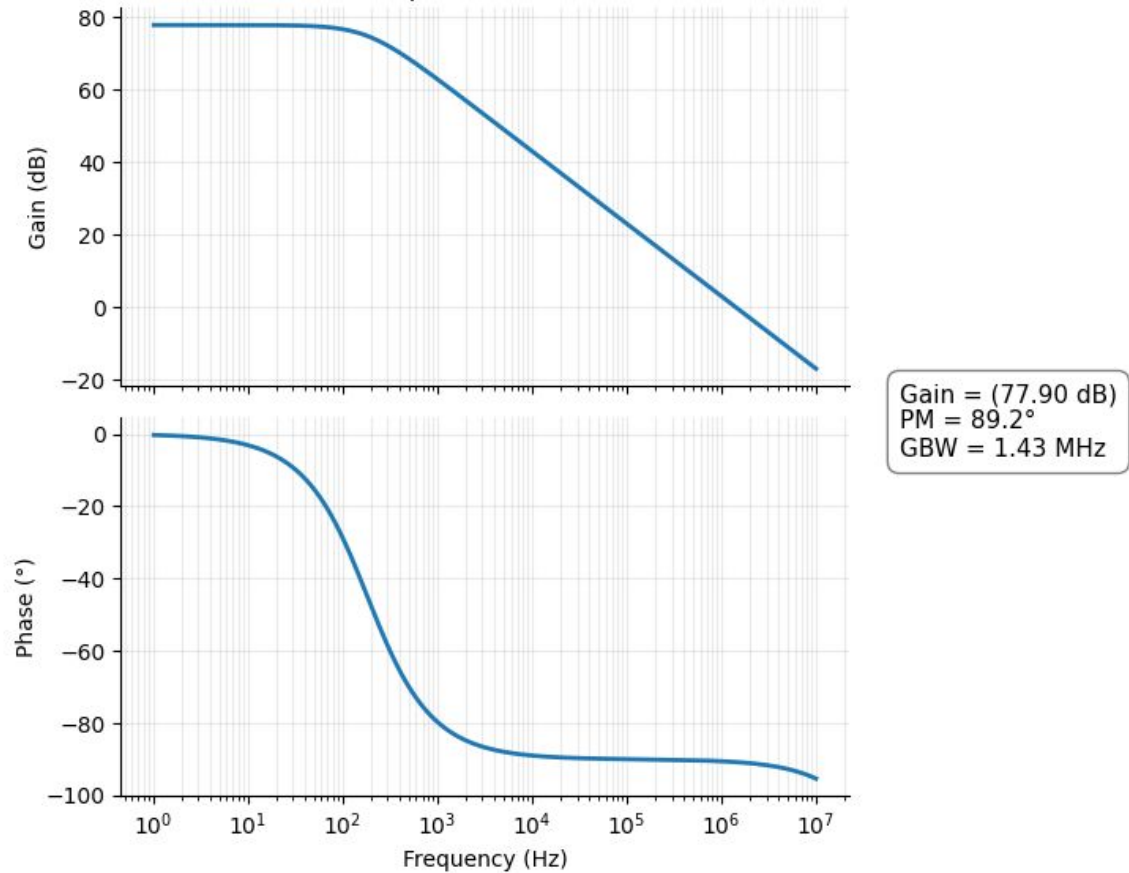
3X Folded Cascode Bode Plot



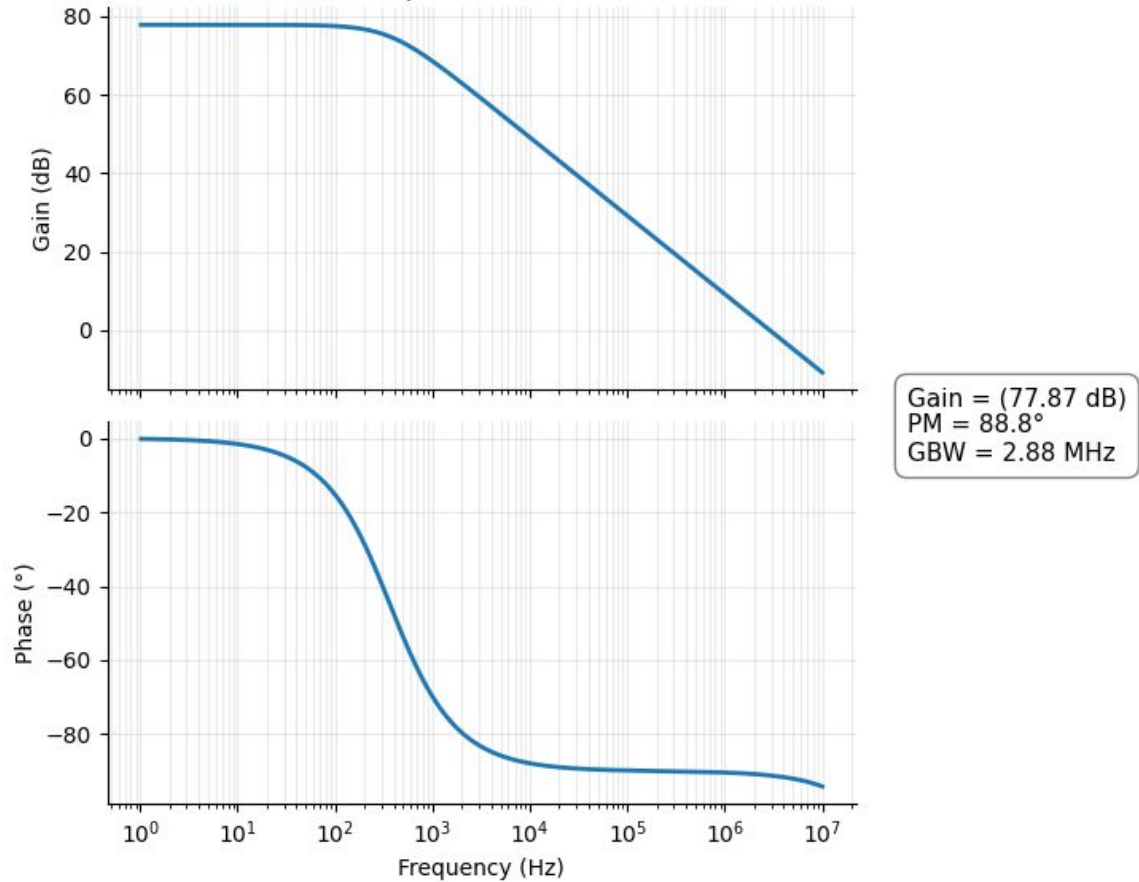
4X Folded Cascode Bode Plot



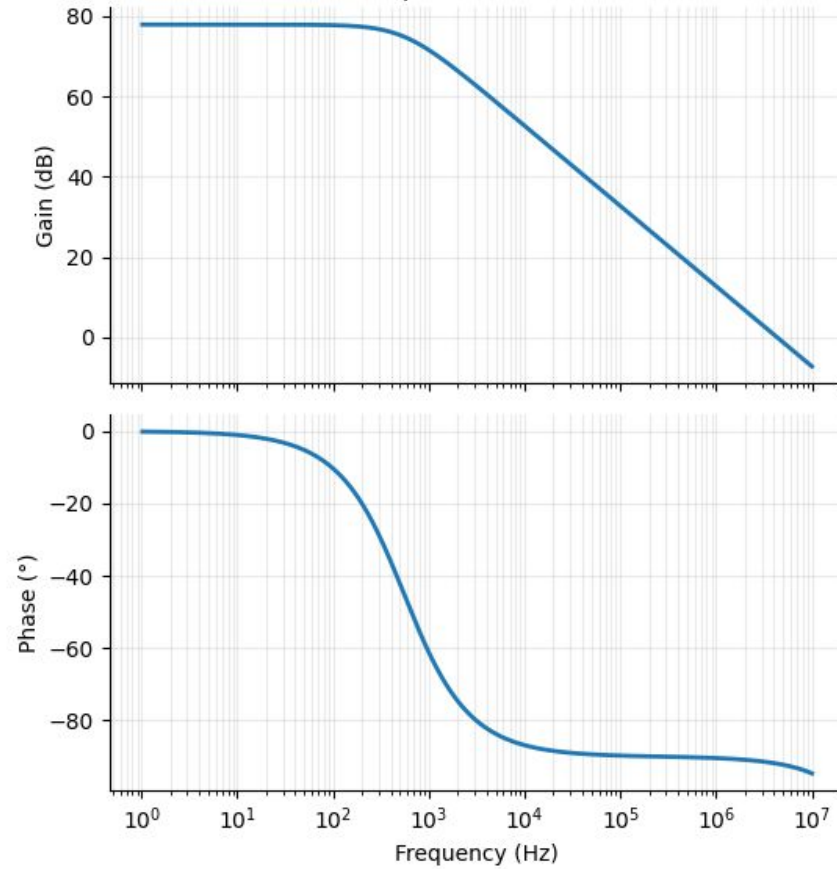
1X Telescopic Cascode Bode Plot



2X Telescopic Cascode Bode Plot

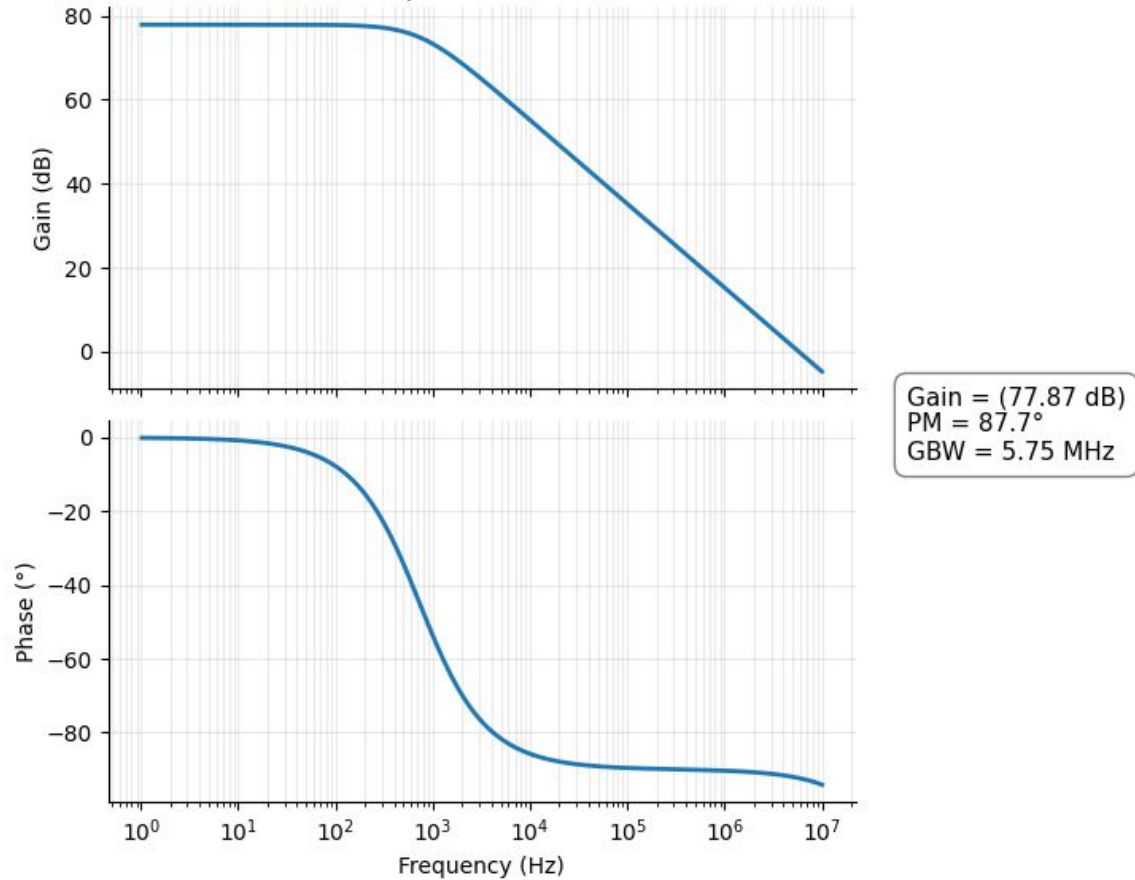


3X Telescopic Cascode Bode Plot



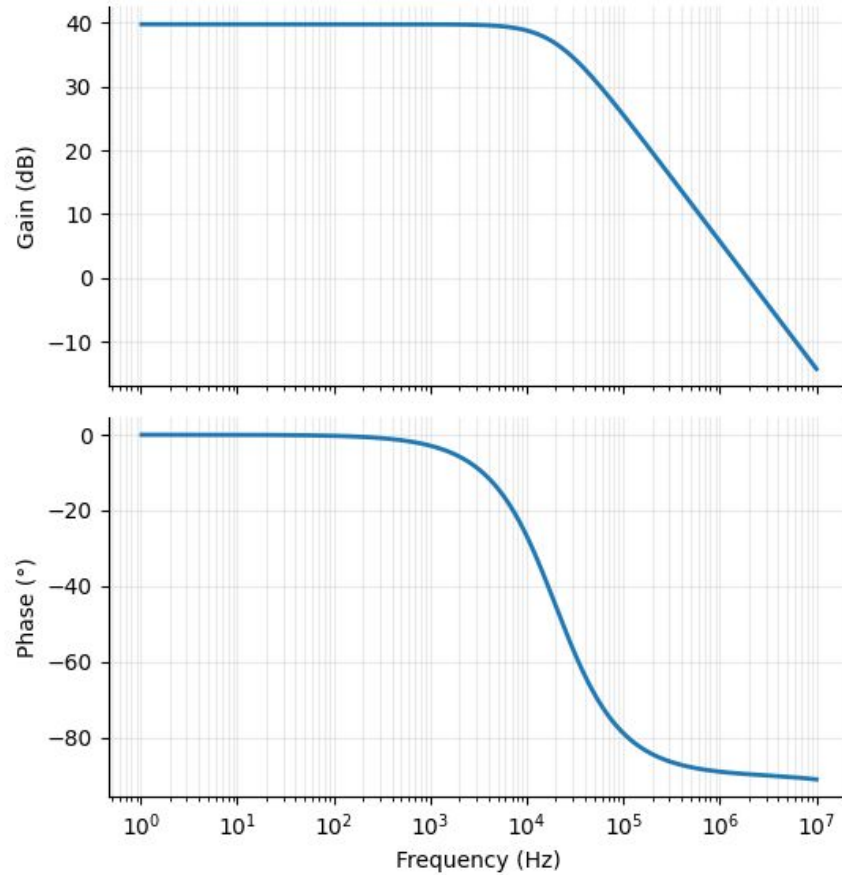
Gain = (77.87 dB)  
PM = 88.0°  
GBW = 4.37 MHz

4X Telescopic Cascode Bode Plot



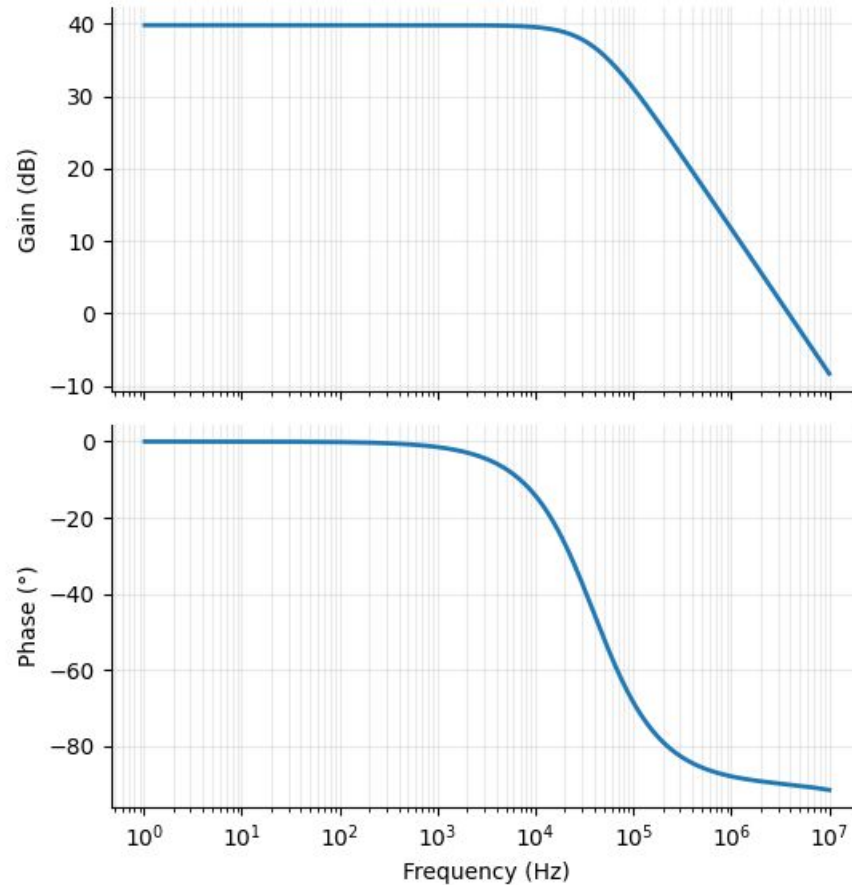


1X Five-Transistor Bode Plot

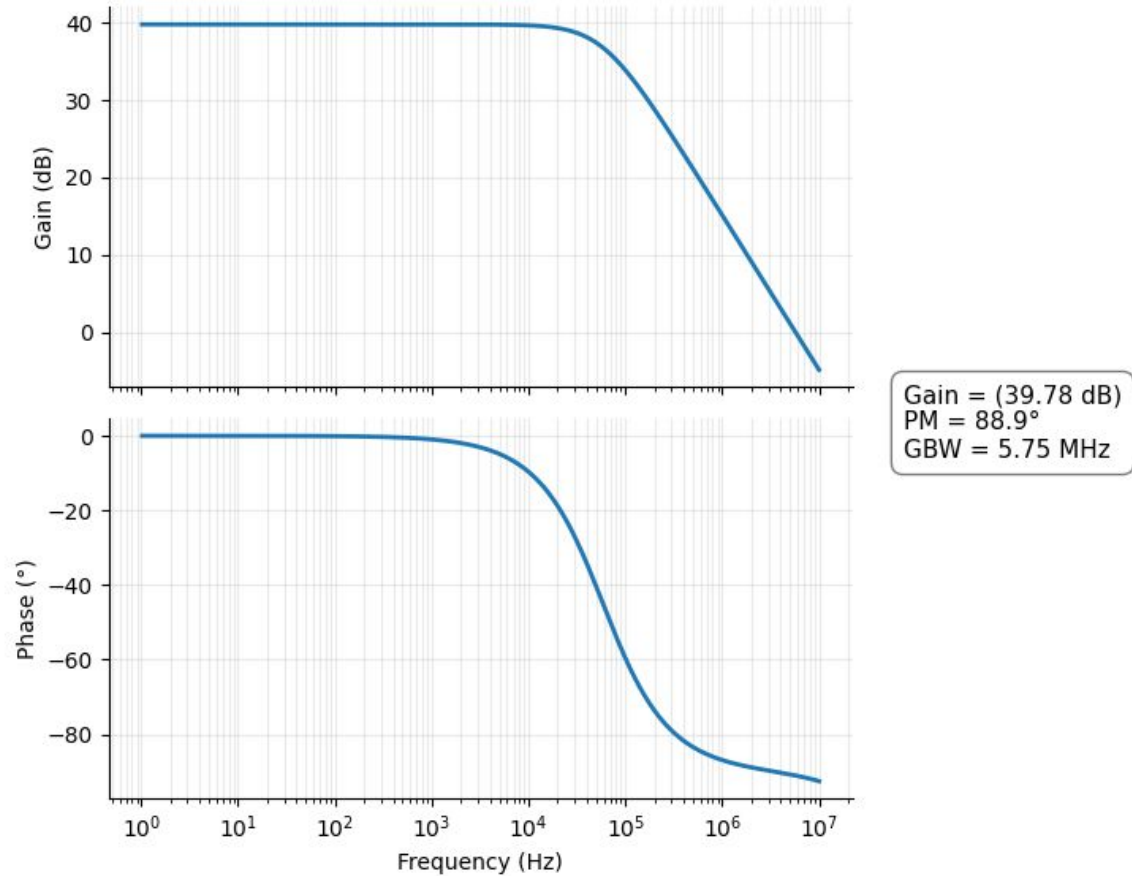


Gain = (39.78 dB)  
PM = 90.4°  
GBW = 1.95 MHz

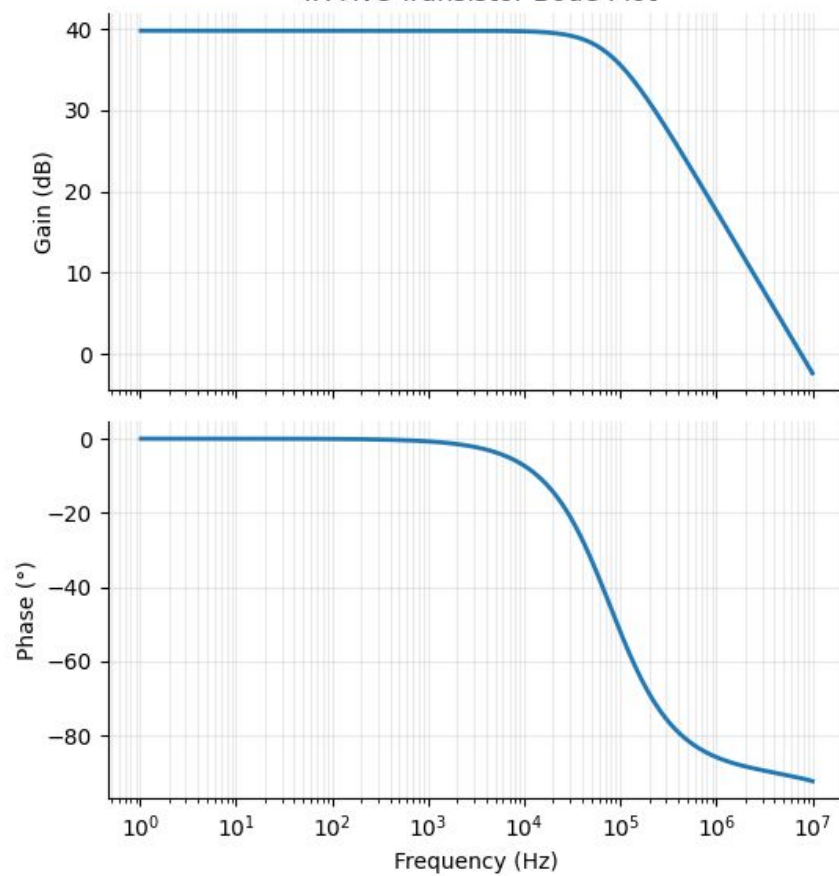
2X Five-Transistor Bode Plot



3X Five-Transistor Bode Plot



4X Five-Transistor Bode Plot

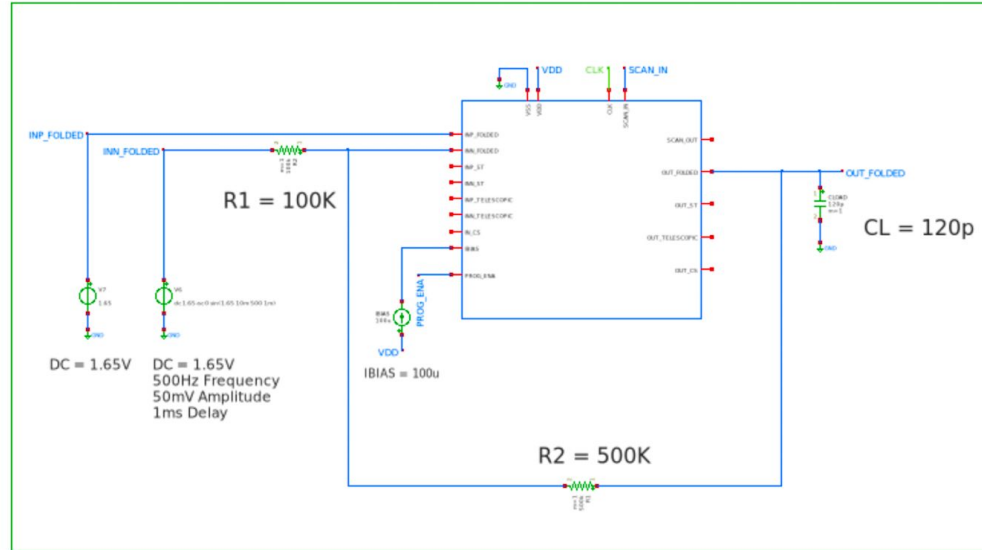


Gain = (39.78 dB)  
PM = 88.5°  
GBW = 7.76 MHz

## Four Topology MOSbius as an Inverter

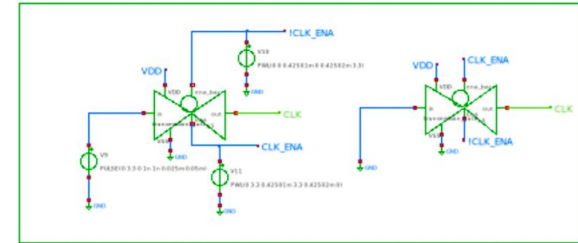
## IMPLEMENTATION

Configure the folded cascode in 4x mode as an inverting amplifier to amplify a sine wave input by 5 while driving a 120pF capacitive load.



## CLOCK GENERATION

Artificially generate a clock signal and ensure it turns off after around .425ms, which is when the scan in signal has fully propagated through the scan chain

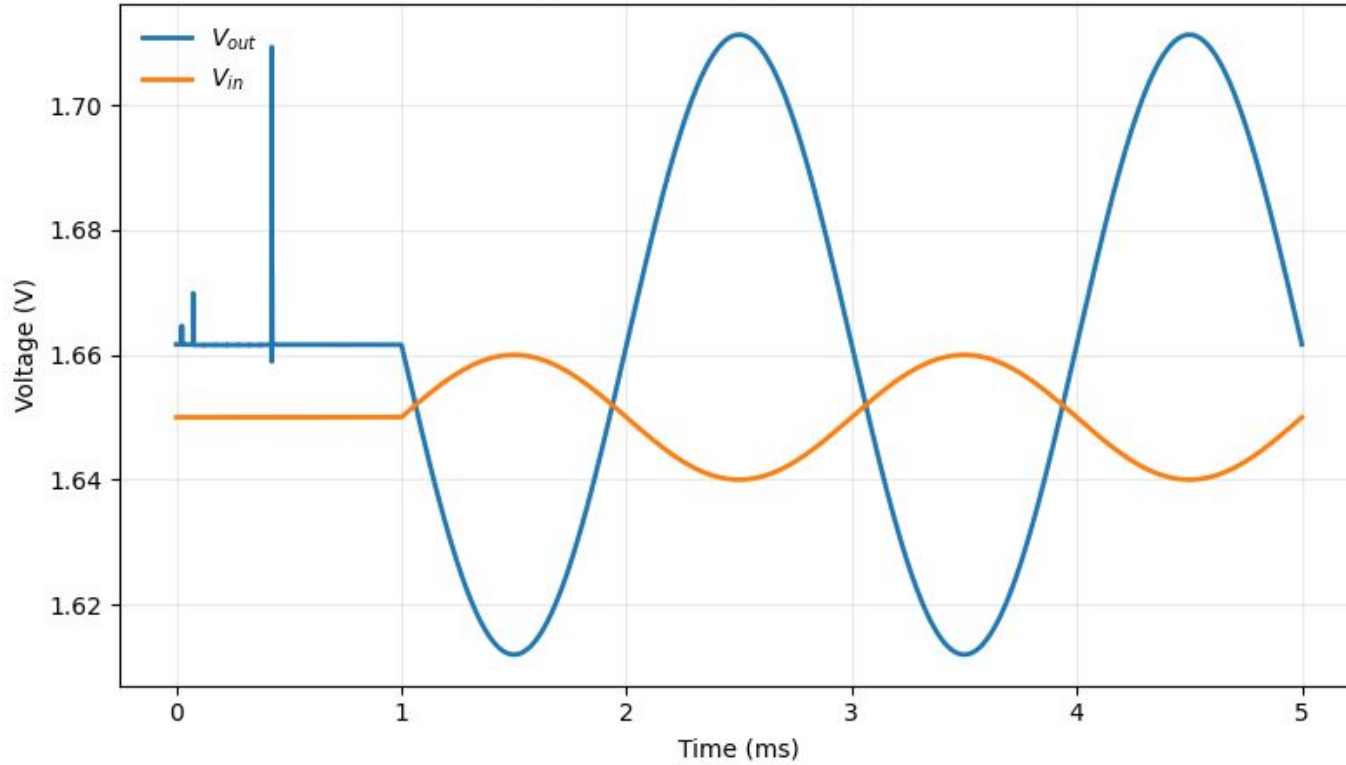


## VDD, GND, SCAN INPUT, PROGRAMMABLE ENABLE

Scan chain input artificially generated to enable folded cascode 4x sizing prog\_ena sets default (low) or program (high)



Folded OTA 2X Transient Response



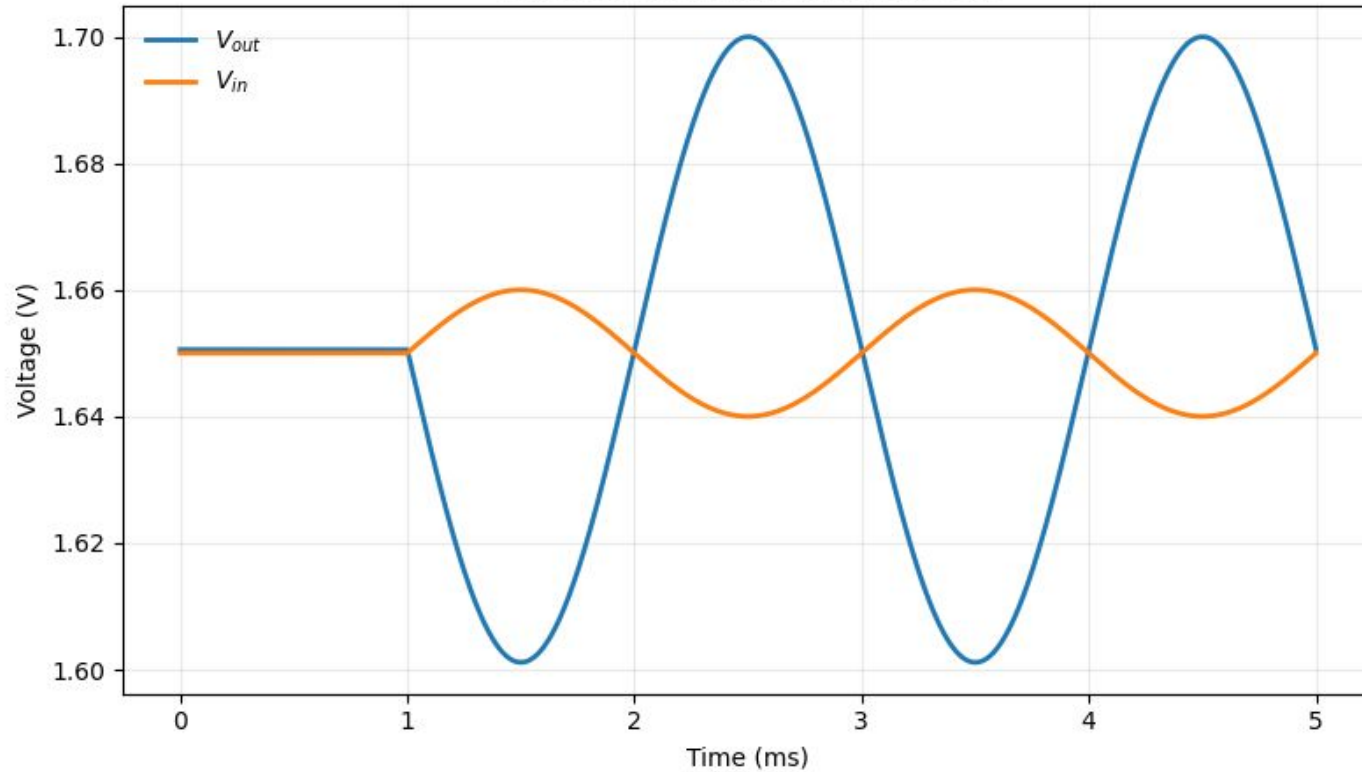
Gain

**4.968728**

V/V

13.92 dB

Telescopic OTA Transient Response



Gain

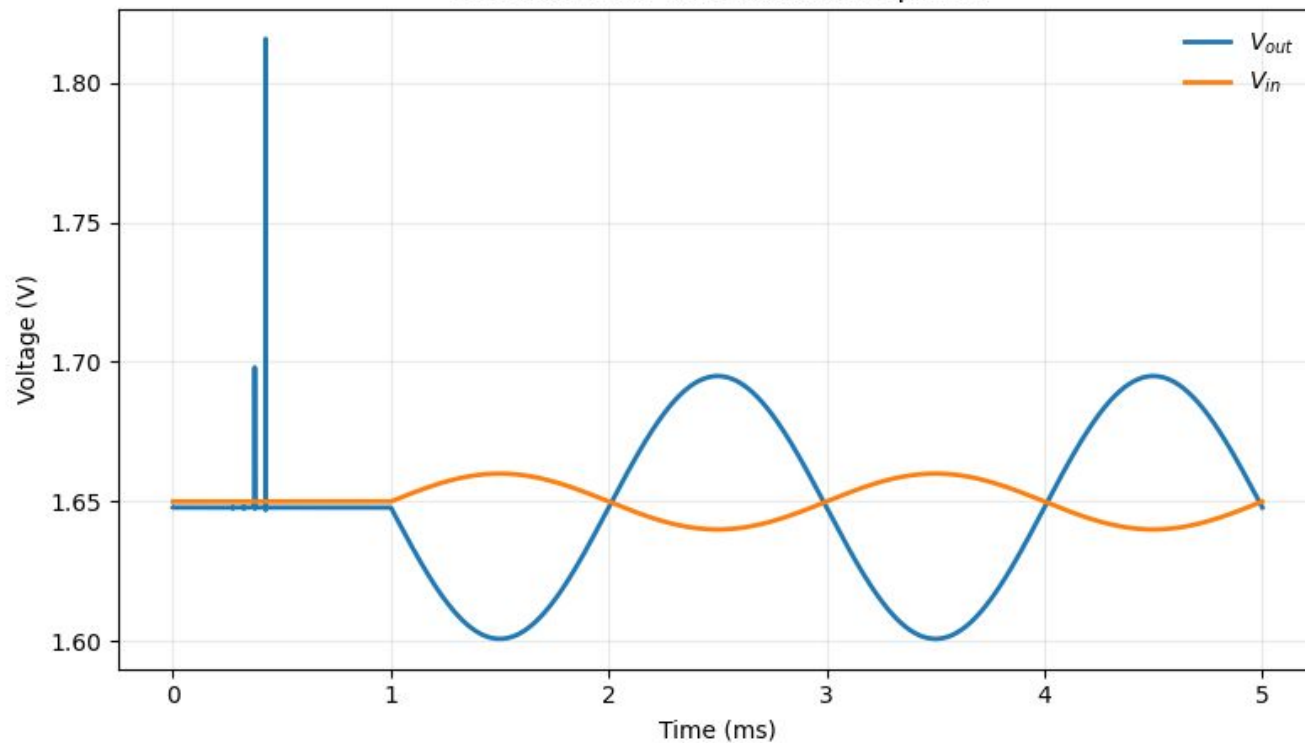
**4.942105**

V/V

13.88 dB



Five Transistor 4X Transient Response



Gain

**4.704538**

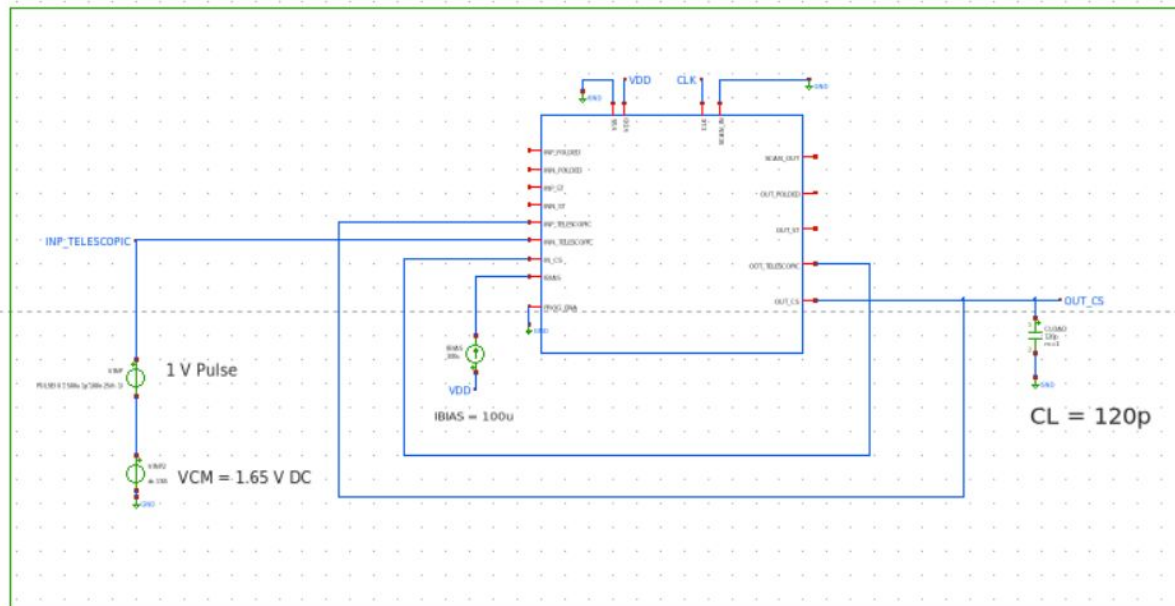
V/V

13.45 dB

## Four Topology MOSbius as a Two-Stage Amplifier

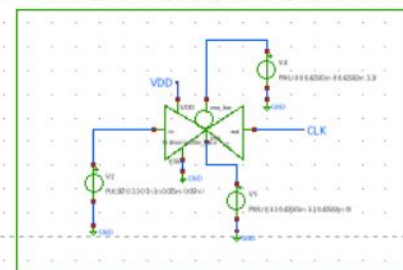
## IMPLEMENTATION

Configure the telescopic cascode in 1x mode and the common source stage in 1x mode as a unity gain amplifier, stepping the input the input by 1 V while driving a 120pF capacitive load without a compensation network.



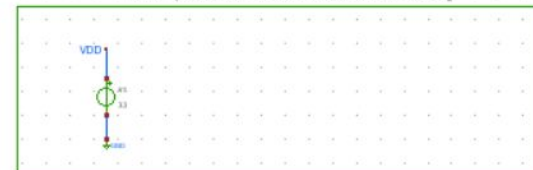
## CLOCK GENERATION

Artificially generate a clock signal and ensure it turns off after around .425ms, which is when the scan in signal has fully propagated through the scan chain

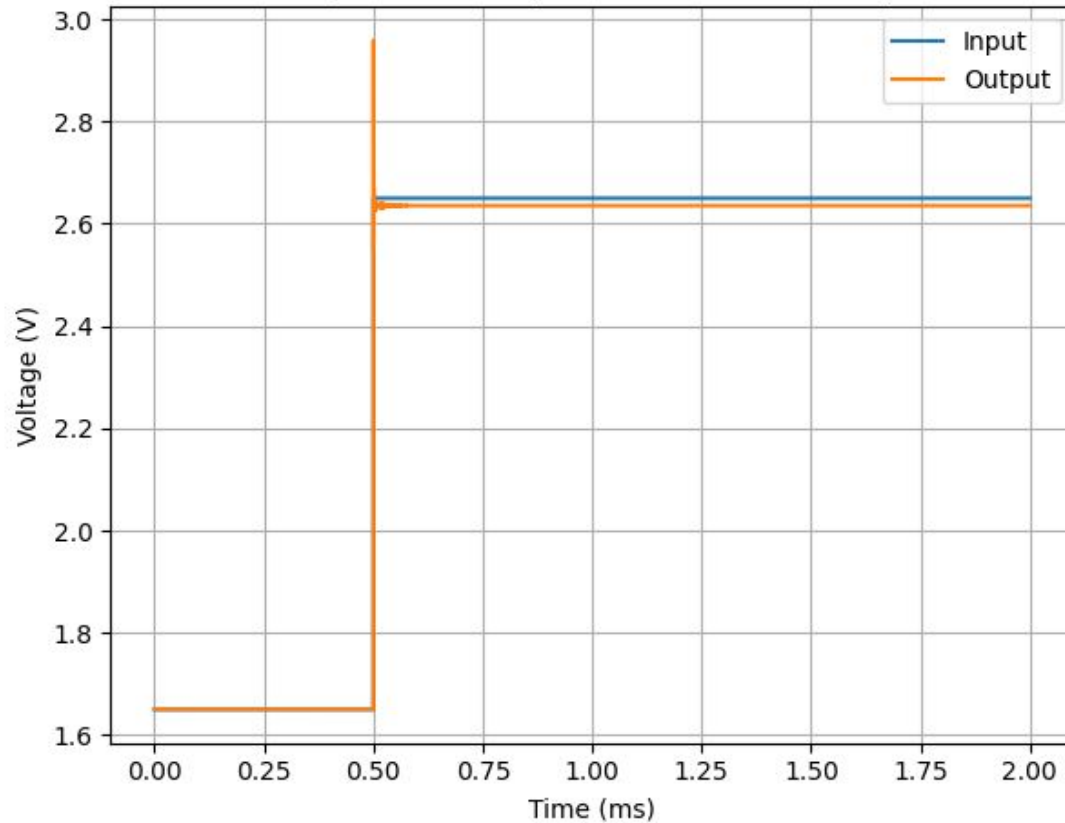


VDD, GND, SCAN INPUT

Scan chain input artificially generated to enable telescopic cascode 1x and common source 1x sizing

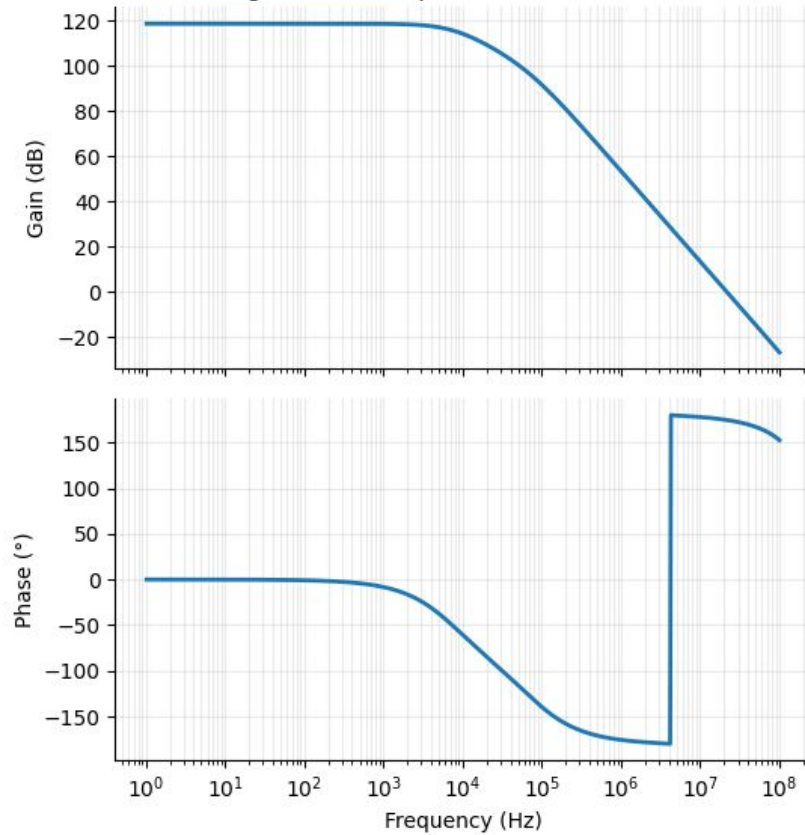


Two Stage 1X Telescopic 1X CS Transient Response



Phase Margin =  $37.16^\circ$

Two-Stage 1X Telescopic Cascode 1X CS Bode Plot



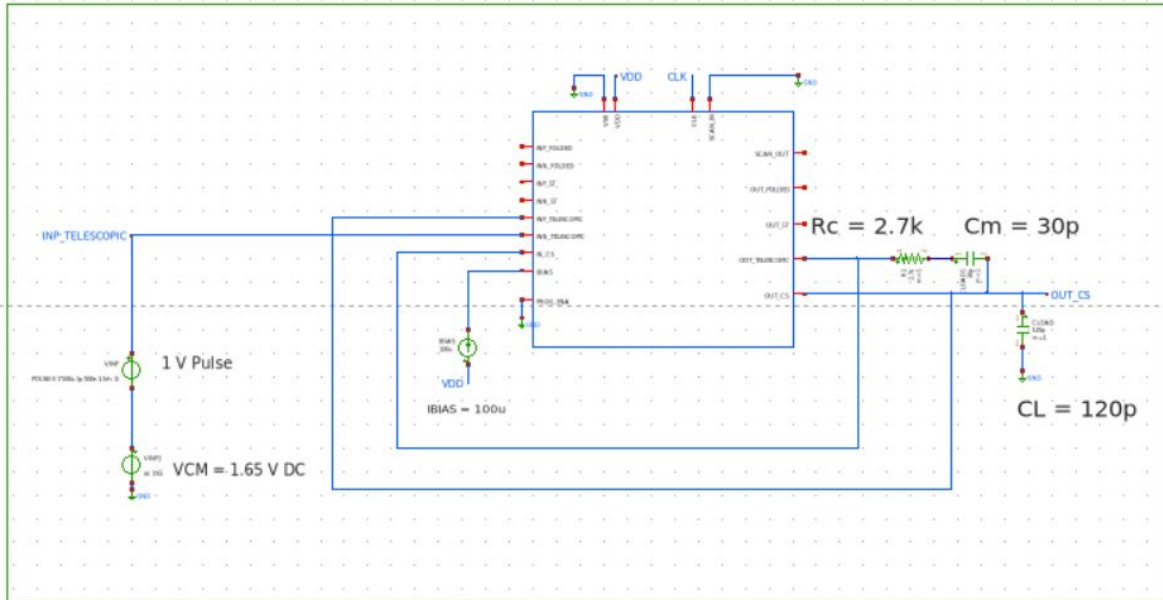
Gain = (118.62 dB)  
PM = 354.6°  
GBW = 21.88 MHz

## Two-Stage Compensation

Users can also implement a Miller capacitor and compensation resistor to stabilize the their amplifier configuration.

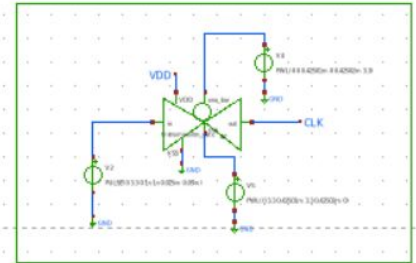
## IMPLEMENTATION

Configure the telescopic cascode in 1x mode and the common source stage in 1x mode as a unity gain amplifier, stepping the input the input by 1 V while driving a 120pF capacitive load without a compensation network.



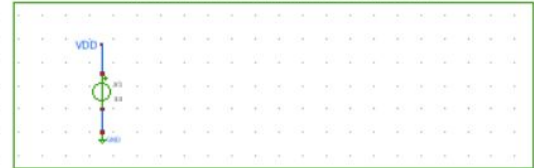
## CLOCK GENERATION

Artificially generate a clock signal and ensure it turns off after around .425ms, which is when the scan in signal has fully propagated through the scan chain

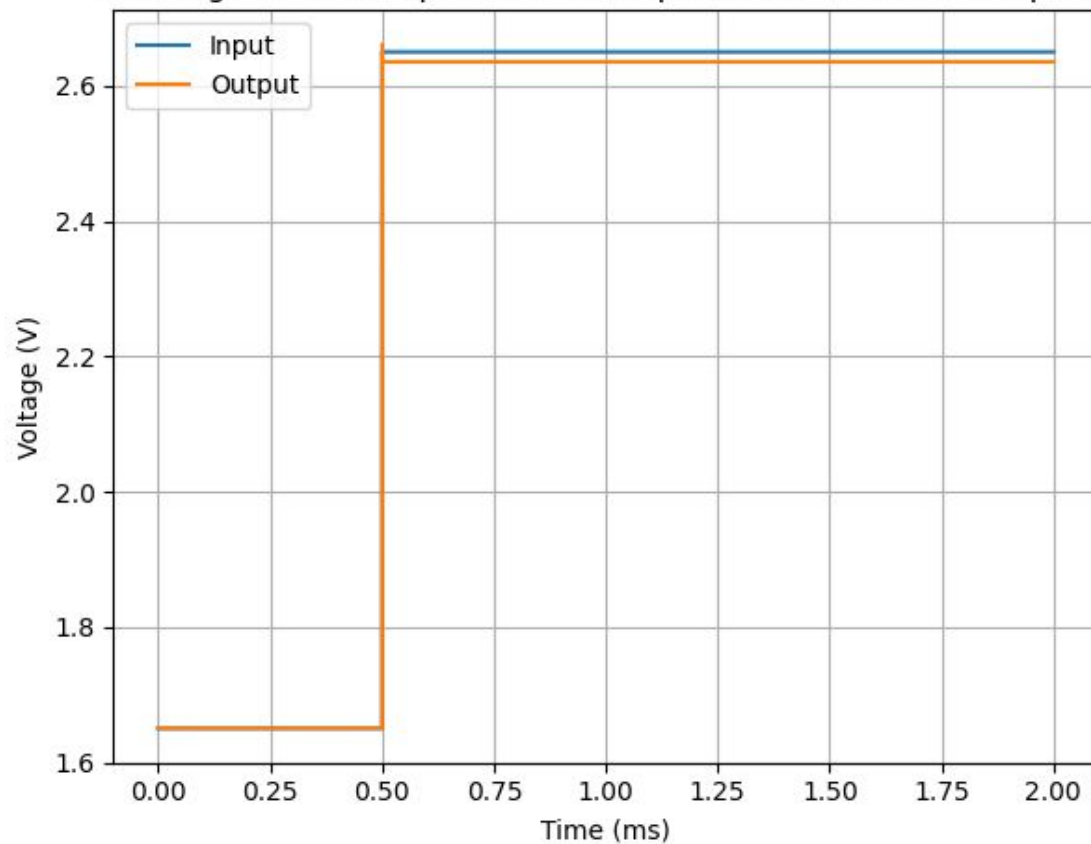


VDD, GND, SCAN INPUT

Scan chain input artificially generated to enable telescopic cascode 1x and common source 1x sizing



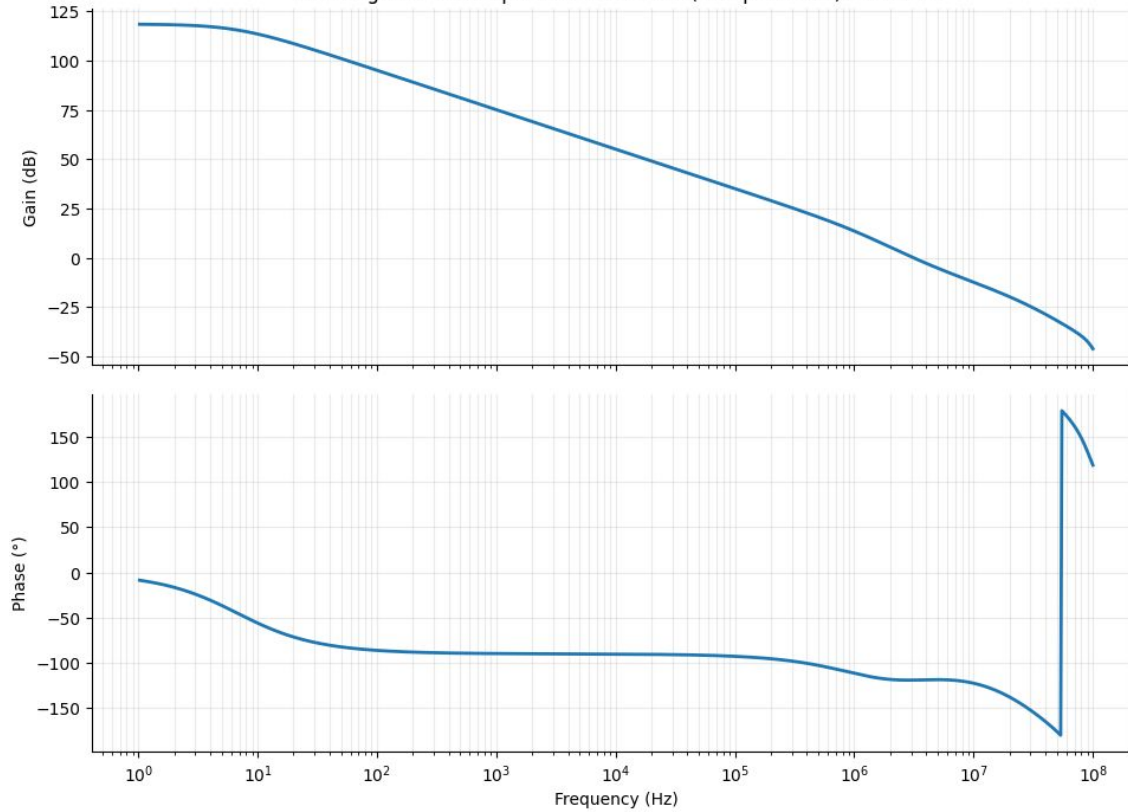
## Two Stage 1X Telescopic 1X CS (Compensated) Transient Response



Phase Margin =  $68.13^\circ$



Two-Stage 1X Telescopic Cascode 1X CS (Compensated) Bode Plot



Gain = (118.52 dB)  
PM = 61.1°  
GBW = 3.24 MHz

## **Full Verification Results:**

### **Folded Cascode:**

<https://docs.google.com/spreadsheets/d/1ste8NNTPVhHXfTPsn5K2P177Z7tUey5MOiy1HPbO30U/edit?usp=sharing>

### **Telescopic Cascode:**

[https://docs.google.com/spreadsheets/d/1UWE3XYKk2p\\_d3-MaK\\_v5p6VSCiXoplRrxV4e-1WU\\_l0/edit?usp=sharing](https://docs.google.com/spreadsheets/d/1UWE3XYKk2p_d3-MaK_v5p6VSCiXoplRrxV4e-1WU_l0/edit?usp=sharing)

### **5-Transistor OTA:**

[https://docs.google.com/spreadsheets/d/12NohRTnEUuWv6hk\\_VIJNx9IA-sGq2unuuVgBkl8fErU/edit?usp=sharing](https://docs.google.com/spreadsheets/d/12NohRTnEUuWv6hk_VIJNx9IA-sGq2unuuVgBkl8fErU/edit?usp=sharing)